# OpenClaw AI Weather Station Tutorial

> Before starting, please complete the following from the `Pre-use Preparation` directory: [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [05-AI Voice Interaction Configuration], and [06-Three Dialogue Configuration Tests]. If you are currently only using TUI / WhatsApp, you can skip `05`. This article assumes the basic environment is already prepared and only covers the AI Weather Station project itself.

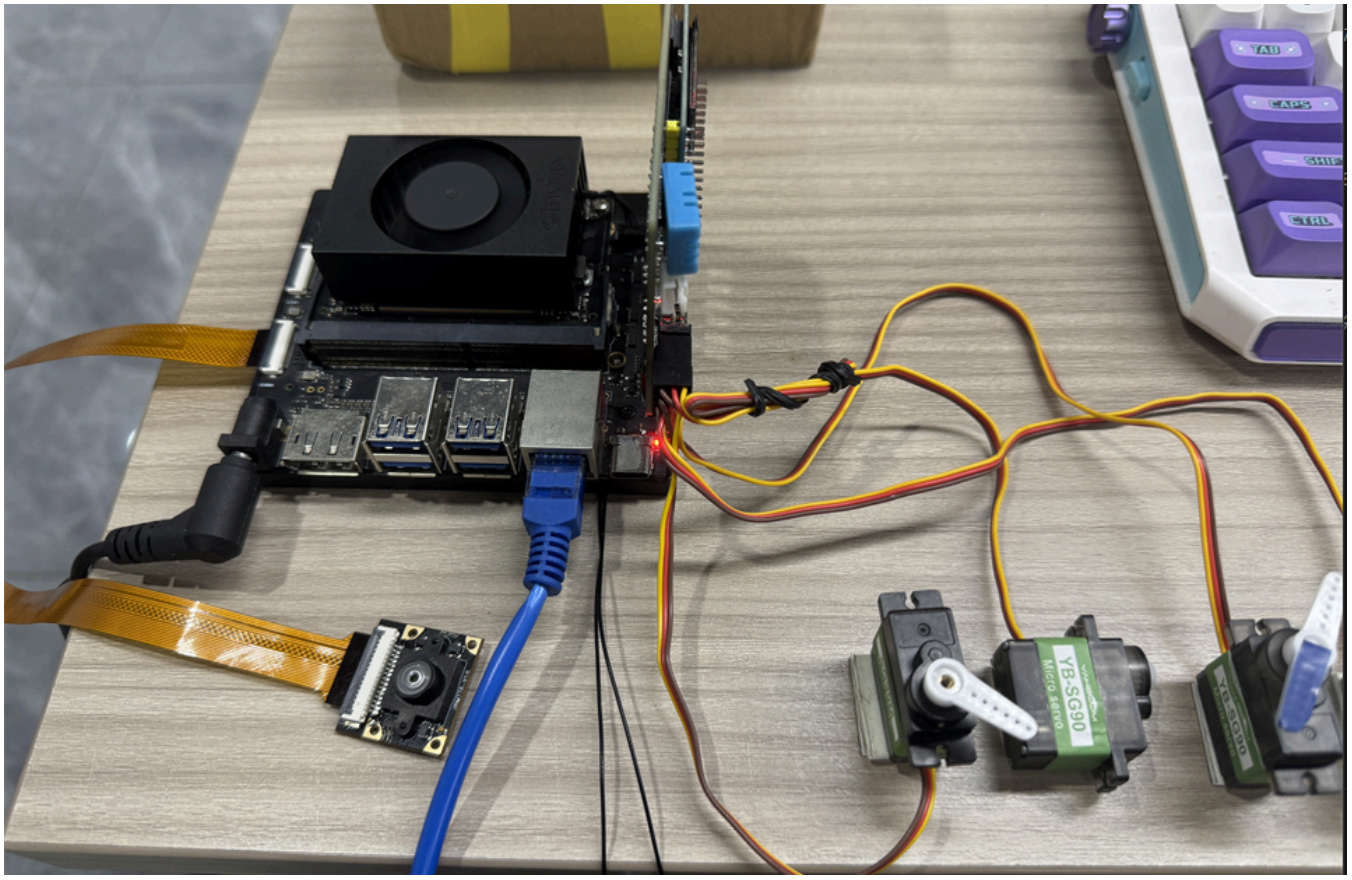
## 1. Tutorial Objectives

This article demonstrates OpenClaw's capabilities in a desktop weather station scenario through temperature/humidity collection, information display, and AI interpretation. After completing this tutorial, readers will be able to:

- Understand how to combine DHT, OLED, and RGB to build a local AI weather station
- Interact with OpenClaw through three methods: WhatsApp, TUI, and voice code
- Complete four project scenarios: environment collection, AI broadcast, weather card, and status alert

## 2. Hardware Preparation

- OpenClaw main control board
- DHT temperature and humidity sensor
- OLED display
- RGB light strip / RGB LED module
- `[Optional] AI voice module + speaker`
- Wiring diagram




# 3. Software Preparation

**Code entry points used in this tutorial**

- Temperature and humidity reading command: `/home/jetson/hardware_hal_demo/dht`
- OLED control command: `/home/jetson/hardware_hal_demo/oled`
- RGB control command: `/home/jetson/hardware_hal_demo/rgb`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`
- Terminal text entry: `/home/jetson/voice_to_openclaw/src/text_main.py`

This tutorial focuses on a "local environment weather station", i.e., using DHT-based indoor temperature and humidity data for display and explanation.

# 4. Testing Project Basic Capabilities

Before starting this project, please complete a basic dialogue self-check using `openclaw tui`; if you haven't done so yet, see [06-Three Dialogue Configuration Tests] first. After passing, proceed to the AI Weather Station specific test.

It is recommended to first enter OpenClaw TUI:

```
openclaw tui
```




Then test the following types of commands in sequence:

- Read current temperature and humidity
- Display weather card on OLED
- If the environment is normal, turn on green light
- Give me a brief environment broadcast

# 5. Three Solutions for Communicating with OpenClaw

| Solution | Suitable Scenarios | Advantages | Recommended Use |
|---|---|---|---|
| WhatsApp | Remote environment viewing, classroom demonstrations | Suitable for remotely asking "how is the environment right now" | Demonstrate "remote weather station assistant" |
| TUI | Local debugging, threshold verification | Easiest for troubleshooting sensor and display logic | First run through the collection pipeline |
| Voice code | Voice broadcast, desktop assistant | Closer to daily usage experience | Use as the final demonstration solution |

**If the optional AI voice module + speaker is selected, Feishu dialogue, TUI, and WhatsApp dialogue need to play back OpenClaw's replies via TTS. You can run the following program in the terminal. The AI voice module dialogue itself does not need to run it again.**

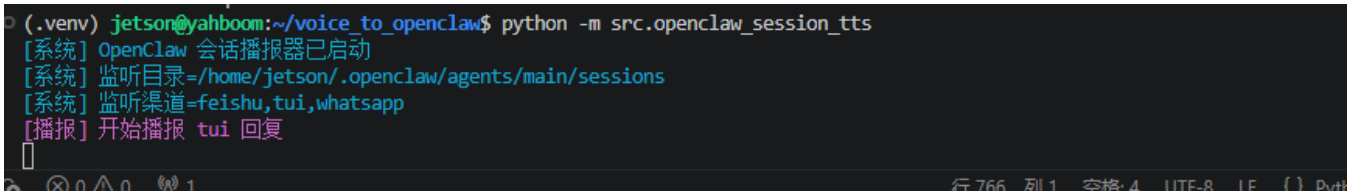
- Modify the TTS broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true  # This parameter controls whether the replies are broadcast. After modification, save it and run the following program in the terminal
```

- Enter the following in the terminal

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```




## 5.1 WhatsApp Solution

For WhatsApp integration and bot creation steps, please refer directly to [06-Three Dialogue Configuration Tests]. After creation, you can remotely view environment status in WhatsApp, for example:

- How's the indoor environment now? If it's too hot, remind me

20:33

How is the indoor environment now? If it is too hot, remind me. 20:34

Current indoor environment readings:

🌡️ **Temperature:** 29.7°C (85.5°F)
💧 **Humidity:** 54%

**Assessment:** The temperature is **just below the hot threshold** (30°C). It's warm but not excessively hot yet.

**Recommendation:**

- The room is comfortably warm, but if the temperature rises above 30°C, I'll remind you with a red light warning
- Current humidity at 54% is in a comfortable range (40-60% is ideal)
- Consider ventilation if you feel warm or if temperature continues to rise

Would you like me to set up a reminder if the temperature exceeds 30°C?

Note: I did not schedule a reminder in this turn, so this will not trigger automatically. 20:34



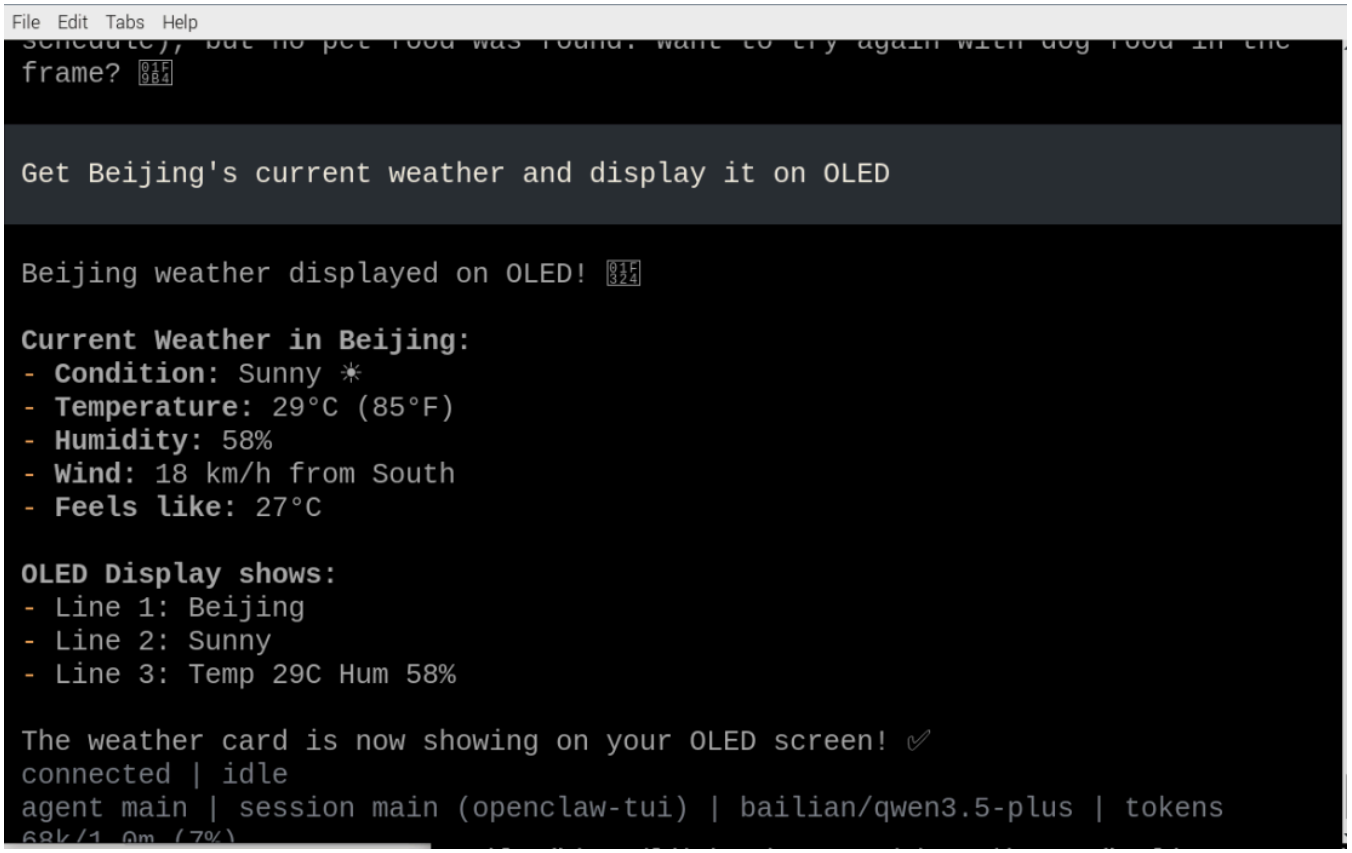
## 5.2 TUI Solution

If you have completed the TUI self-check in article 0, directly enter terminal text dialogue:

```
openclaw tui
```

Then enter project-related commands, for example:

- Get current weather for Beijing and display it on OLED

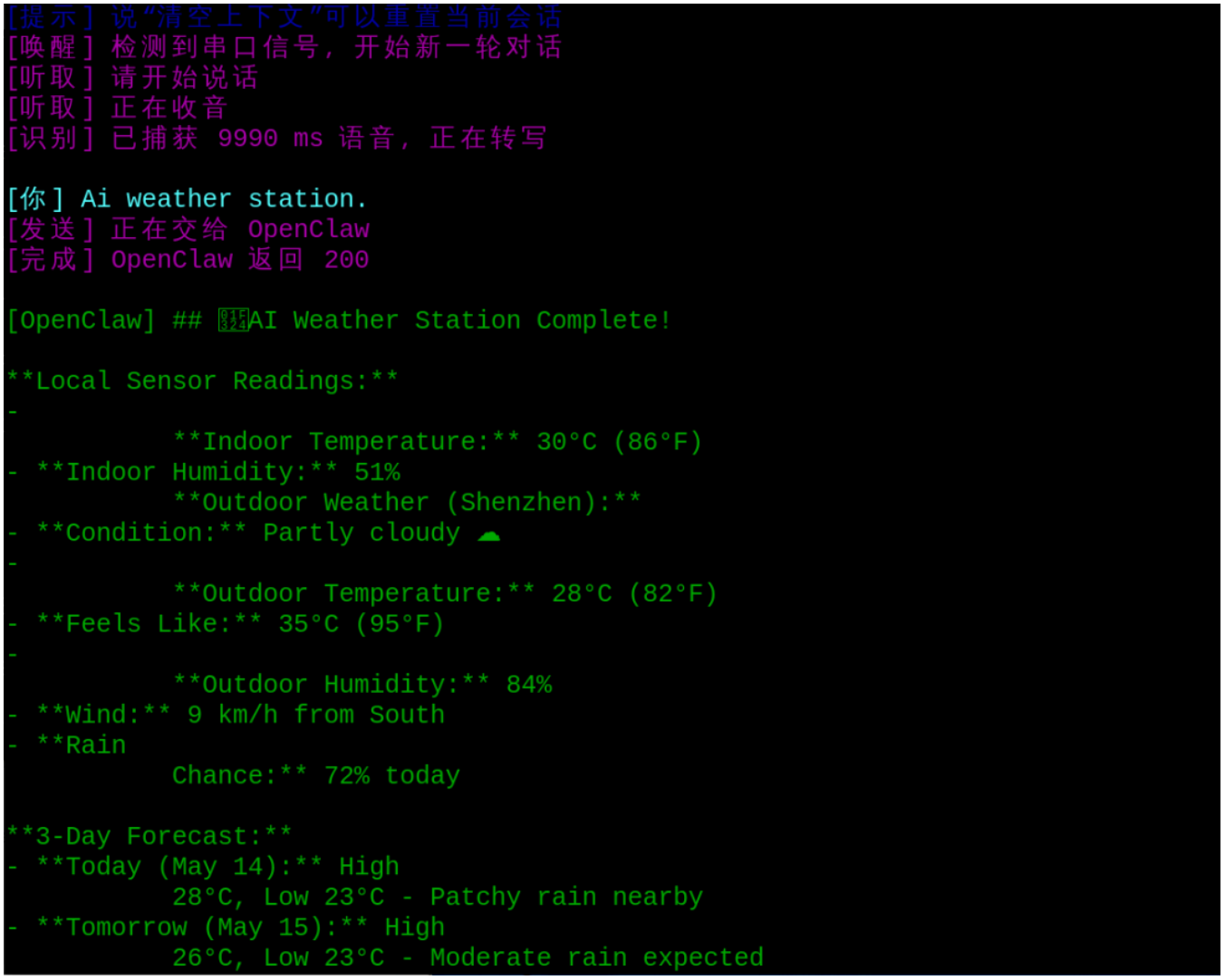



## 5.3 Voice Interaction Solution

```
Requires optional AI voice interaction module + speaker
```

For voice module installation, environment configuration, and basic wake-up process, please refer directly to [05-AI Voice Interaction Configuration]. After wake-up, you can directly speak control commands, for example:

- AI weather station




# 6. Project Implementation

Note: The following demonstrations are all conducted using `openclaw tui` dialogue. You can also choose WhatsApp or voice methods to complete them.

## 6.1 Scenario 1: Environment Collection

### Experiment Objective

Read the local ambient temperature and humidity to complete the most basic data sampling for the weather station.

### Effect Description

OpenClaw will call DHT to get the current environment data and return the results to the user, providing a real-time "weather reading" of the indoor environment.

### Example Commands

- Read current ambient temperature and humidity
- What's the current indoor temperature and humidity
- Help me check the current weather data

## 6.2 Scenario 2: AI Broadcast

### Experiment Objective

Make OpenClaw not just report numbers, but also summarize the environment status like a weather broadcaster.

### Effect Description

OpenClaw can summarize "26 degrees, 57% humidity" into more natural phrases, such as "the environment is relatively comfortable", "it's a bit stuffy", "the air is slightly dry", etc.

### Example Commands

- Give me a brief environment broadcast
- Is the environment comfortable now
- Make a weather summary based on current readings

## 6.3 Scenario 3: OLED Weather Card

### Experiment Objective

Display local temperature and humidity results in a "weather card"-style page.

### Effect Description

In addition to plain text results, OLED can also arrange location, status, temperature, and humidity into a compact card, making the entire project more like a desktop display device.

### Example Commands

- Display an indoor weather card on OLED
- Get current weather for xx location and display it on OLED

## 6.4 Scenario 4: RGB Status Alert

### Experiment Objective

Use RGB colors to quickly indicate the current environment status.

### Effect Description

For example, turn on the green light when the environment is comfortable, yellow light when it's hot, red light when abnormal. This way, even without reading text, you can quickly know the current environment status.

### Example Commands

- If the environment is normal, turn on green light
- If the temperature is high, turn on yellow light
- If abnormal, flash red light